

DexOS System Architecture & Feature Audit

This document outlines the current state, codebase structure, database schema, and UI screens for the DexOS application (formerly Zyrbit). Use this as a comprehensive guide for AI context/research to rebuild and refactor the app into a premium, unified AI-powered personal operating system.

1. System Overview & Tech Stack

DexOS is designed as a personal operating system bridging productivity, wellness, finance, and learning under a central gamified economy and an AI command layer.

Core Technology Stack

- % Frontend: React (v19) + Vite (v8) + TailwindCSS (v4)
- % Database & Auth: Supabase (PostgreSQL client)
- % AI Integration: Gemini API (gemini-2.5-flash) via developer key
- % Visual Assets: Scalable SVGs, custom ambient gradients, clean pitch-black background (#121214), neon typography
- % PWA Wrapper: Workbox PWA configuration with manifest, custom install prompts, offline service worker, and mobile-responsive styling.

2. Directory & Codebase Structure

The active production files are organized as follows:

zyrbit/	
%%% index.html	# Entry page with initial
%%% package.json	# Project scripts and depe
%%% vite.config.js	# Vite configurations + PW
%%% public/	
% %%% manifest.json	# PWA manifest configurati
% %%% app-icon.svg	# Main SVG logo asset (sla
% %%% favicon.svg	# Browser tab icon
% %%% icons/	# Generated PNG icons for
% %%% qa_screenshots/	# Current visual screensho
%%% src/	
% %%% main.jsx	# React root hydration
% %%% App.jsx	# Main router, auth state
% %%% index.css	# Base theme variables, ut
% %%% design/	
% %%% theme.css	# Dark mode tokens (accent
% %%% typography.css	# Space Grotesk + Inter fo
% %%% context/	
% %%% SubscriptionContext.jsx	# Premium subscription
% %%% hooks/	
% %%% useHabitReminders.js	# PWA notifications schedu

%	%	%%%	useInstallPrompt.js	#	Custom browser PWA banner
%	%%%	lib/			
%	%	%%%	supabase.js	#	Supabase client instantiation
%	%	%%%	gemini.js	#	Gemini API wrapper (task)
%	%	%%%	gravity.js	#	Gravity Score algorithm
%	%	%%%	ranks.js	#	Gamification rank bracket
%	%	%%%	zyrons.js	#	Gamified Zyrons ledger
%	%%%	pages/			
%	%	%%%	Zenith.jsx	#	Main Life OS Dashboard
%	%	%%%	Growth.jsx	#	Study hub & Project tracker
%	%	%%%	Health.jsx	#	Vitals, sleep, exercise
%	%	%%%	Wealth.jsx	#	Monthly budgets, expense
%	%	%%%	Dex.jsx	#	Jarvis AI Voice & Chat A
%	%	%%%	Stats.jsx	#	Performance charts & his
%	%	%%%	Profile.jsx	#	User rank, transaction l
%	%	%%%	Challenge.jsx	#	Sprints / habit challeng
%	%%%	components/			
%		%%%	Logo.jsx	#	Reusable ZyrbitMark & Zy
%		%%%	BottomNav.jsx	#	Mobile persistent naviga
%		%%%	FeedbackWidget.jsx	#	Floating bug reporter
%		%%%	GravityRing.jsx	#	Concentric ring visualiz
%		%%%	InstallBanner.jsx	#	Sticky PWA install drawe
%		%%%	...	#	Specialized sub-widgets

3. Database Schema

The backend runs on Supabase (Postgres). The active tables and migrations are detailed in Zyrbit_Full_Schema.sql and dexos_v1_migration.sql.

Key Tables

- 1. profiles: User metadata, username, avatar, premium status.
- 2. habits: User routine definitions (title, frequency, difficulty, zone: mind/body/growth/soul).
- 3. habit_logs: Logs daily completion events, skips, or notes.
- 4. focus_sessions: Tracks focus timer blocks (duration_minutes, subject, tags).
- 5. expenses: Financial transactions (amount, category, date, type: income/expense).
- 6. health_logs: Tracks daily metrics: sleep hours, water intake (ml), weight, active calories.
- 7. goals: Tracks user-configured targets (target_value, current_value, deadline).
- 8. zyrons_ledger: Logs every XP/currency transaction (rewarded for habits, focus, budget consistency).

4. UI/UX Breakdown by Tab (with Screenshots)

Here is a detailed audit of every active screen in the system, referencing the actual screenshot files generated during testing.

A. Splash & Entry Flow

- % File: src/screens/SplashScreen.jsx & index.html (pure HTML splash)
 - % Aesthetics: Charcoal-black surface (#121214) featuring a central white Zyrbit icon with a glowing pulse animation.
 - % Visual Ref: public/qa_screenshots/1_splash.png
 - % Functionality: Checks for user session tokens. If anonymous or unauthenticated, redirects to LoginScreen. If authenticated, fades out and loads the primary shell.
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B. Onboarding & Setup

- % Files: src/screens/OnboardingScreen.jsx & src/screens/GoalSetupScreen.jsx
 - % Aesthetics: Glassmorphic onboarding cards sliding horizontally, showcasing the Core Pillars (Mind, Body, Wealth, Growth).
 - % Visual Ref: public/qa_screenshots/2_onboarding.png & public/qa_screenshots/4_goal_setup_screen.png
 - % Functionality: Guides new users to seed initial goals (e.g. targeting monthly savings, active workout days per week, and sleep hours) to configure the database defaults.
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C. Zenith Dashboard (Home Tab)

- % File: src/pages/Zenith.jsx
 - % Aesthetics: Dark premium workspace layout featuring a header with the central Gravity Score Ring (src/components/GravityRing.jsx). Displays alert banners at the top if scores drop too low.
 - % Visual Ref: public/qa_screenshots/5_zenith_page.png & public/qa_screenshots/7_zenith_with_new_habit.png
 - % Widgets Included:
 1. Gravity Ring: Centered visual showing a score from 0 to 100 (computed from habit consistency, sleep, and budget).
 2. OS Warnings System: Collated warnings list (e.g., Capacity Alert: Recovery depleted or Runway Alert: low financial buffer).
 3. Routines / Habit List: Segmented list by time of day (Morning/Afternoon/Evening) allowing one-tap completion or skip logs.
 4. AI Operating Briefing: A dynamic greeting block summarizing performance trends using Gemini API.
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D. Growth Engine (Growth Tab)

- % File: src/pages/Growth.jsx
 - % Aesthetics: Slate-grey cards with neon-blue accents. Divided into sub-tabs: Today, Projects, and Sprints.
 - % Visual Ref: public/qa_screenshots/9_growth_page.png
 - % Widgets Included:
 1. Focus Session Timer: A custom Pomodoro circle timer with toggle states (Work/Short Break/Long Break). Integrates a Blackout Mode option to lock tabs or hide distractions.
 2. Projects Tracker: Lists active projects, milestones, task completion percentage, and estimated hours spent.
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E. Health Telemetry (Health Tab)

- % File: src/pages/Health.jsx
 - % Aesthetics: Forest-green and dark teal color scheme. Standardized card layouts for health signals.
 - % Visual Ref: public/qa_screenshots/15_health_page.png
 - % Widgets Included:
 1. Vitals Grid: Inline quick displays showing Sleep quality (hours logged vs target), active calories, and current weight.
 2. Water Card: Incremental logger allowing the user to tap +250ml or +500ml options with animated waves.
 3. Dexos Recovery Score: A daily score computed based on sleep hours and exercise level.
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F. Wealth Tracker (Wealth Tab)

- % File: src/pages/Wealth.jsx
 - % Aesthetics: Emerald green highlights with graphite background surfaces.
 - % Visual Ref: public/qa_screenshots/17_wealth_page.png
 - % Widgets Included:
 1. Financial Runway: Computes runway buffer in months based on total savings divided by monthly average burn rate.
 2. Expense/Income Logs: A fast-entry ledger split by category (Food, Housing, Subscriptions, Investment).
 3. Mirror Bar: Displays a progress bar comparing total spent this month against the designated budget.
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G. Dex AI (Jarvis Tab)

- % File: src/pages/Dex.jsx
 - % Aesthetics: Holographic blue/cyan glowing waves. Implements a large central AI orb widget that expands/contracts based on response states.
 - % Visual Ref: public/qa_screenshots/19_dex_page.png & public/qa_screenshots/21_dex_summary_tab.png
 - % Widgets Included:
 1. Chat Interface: Realtime conversational interface with the personal operating assistant. Includes options to fetch daily briefings or execute rapid tasks.
 2. Voice Recognition: Utilizes Web Speech API for voice commands (e.g. "Log 500ml of water", "Log a 45 min focus session").
 3. Knowledge/Quiz Mode: Generates interactive quick quizzes based on the user's logged focus session subjects.
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H. Profile & Wallet (Profile Tab)

- % File: src/pages/Profile.jsx
 - % Aesthetics: Gold and silver rank overlays. Dark chrome interfaces.
 - % Visual Ref: public/qa_screenshots/23_profile_page.png
 - % Widgets Included:
 1. Rank Banner: Displays current tier (e.g. Ascendant, Strategist) based on lifetime Zyrons.
 2. Zyrons Transactions: Lists a detailed history of XP earnings (ledger).
 3. Cosmos Shop: Conceptual store showing unlockable features or premium themes.
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5. Claude Blueprint for Rebuilding DexOS (v2)

When instructing Claude or another LLM to refactor this codebase into DexOS v2, provide the following blueprint:

Phase 1: Core Cleanups & Routing

- % Action: Consolidate the router in App.jsx to map only the active pages (Zenith, Growth, Health, Wealth, Dex, Stats, Profile). Remove unrouted files like DailyLog.jsx, Dashboard.jsx, Journal.jsx, and Orbit.jsx.
- % Goal: Simplify navigation paths, making page changes snappy. Keep the bottom bar navigation limited to the core pillars.

Phase 2: Schema Hardening

- % Action: Ensure that postgres logs are correctly structured. Reconcile the fields for focus durations (ensure database is aligned to duration_minutes or duration_mins, not a mix of both).
- % Goal: Prevent silent database errors during logs.

Phase 3: AI Service Layer Consolidation

- % Action: Extract the API calls from Dex.jsx and Zenith.jsx into a unified helper file (e.g. src/lib/gemini.js). Implement structured JSON output formats using schemas to ensure the AI summary widget always parses correctly.
- % Goal: Eliminate duplicate fetch code, establish global error boundaries for the AI.

Phase 4: Standardize Theme Variables

- % Action: Ensure styling relies on CSS variables defined in src/design/theme.css (--color-success, --color-accent, etc.) instead of hardcoded hex values in component style props.
- % Goal: Allow unified theme toggles to propagate instantly to all sub-components.